

IGB100 Cycle 1 Mini-Game Design Document

Annie's Last Stand

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Game Overview

Annie's Last Stand is a top-down shoot 'em up (SHMUP) where players defend the world's final nuclear silo against relentless waves of autonomous mech enemies. Players assume the role of Annie, a lone elite operative tasked with holding the line until the silo completes its launch sequence. The narrative setup draws from modern crisis scenarios, grounding the conflict in a context that feels urgent and relevant to the player, increasing emotional investment from the outset.

The intended player experience centres on tension and urgency, evolving into strategic mastery as players gain access to upgrades. Gameplay follows a loop of high-intensity combat and tactical decision-making, with continuous 360-degree movement and shooting. Players must prioritise threats such as fast swarm units, heavily armoured siege mechs, and aerial drones that bypass defences, creating constant pressure and engagement.

Defeated enemies drop "Scrap," which can be spent on weapon upgrades, turret deployment, and wall repairs. This introduces a meaningful risk-reward dynamic, forcing players to balance immediate survival with long-term defensive investment. The pacing, enemy variety, and upgrade systems are carefully tuned to maintain flow, challenge, and player agency, encouraging a satisfying sense of mastery over repeated playthroughs.

Annie herself is designed as an attractive, anime-styled character with a balance of visual appeal and practical armour. Her design contributes to player engagement by fostering attachment and identity, providing an immediate visual hook, and reinforcing the player's investment in the unfolding experience. This character-driven appeal works in concert with gameplay tension, enhancing retention and deepening the emotional connection to both the action and the stakes.

The game is set in a confined industrial courtyard with a gritty sci-fi aesthetic inspired by *Nier: Automata*, *Honkai Impact*, and *Towa no Yuugure*. Visual clarity is emphasised through colour-coded threats (blue for player, red for enemies), while the overall design reinforces the desperate, high-stakes "last stand" atmosphere, keeping players immersed in both the mechanical and emotional experience.

Week 1: SHMUP Analysis & PX Goals

Analysis of *Galaga* revealed that constrained movement and one-hit death mechanics created a strong sense of fragility and tension, making each movement decision meaningful.

In contrast, *Vampire Survivors* demonstrated how automated combat systems shift player focus toward positioning and survival, producing a feeling of progressive empowerment as the player scales in strength.

From studio discussions, three core PX goals were identified:

- Urgency
- Strategic Mastery
- Tension

This project primarily focuses on Tension during combat and Strategic Mastery through the upgrade system.

Week 2: Concept Development & Revision

The initial concept proposed a squad-based system involving AI-controlled allies. However, feedback identified that managing multiple agents would dilute the core SHMUP experience, particularly the emphasis on precise movement and dodging.

The design was revised to a single-player “lone operative” model, with strategic depth maintained through deployable turrets and defensive structures. This shift strengthened player agency and reinforced Annie as the central actor, while preserving the defensive gameplay theme.

Inspirations include defensive scenarios such as *Helldivers* missions and large-scale last-stand battles like the Battle of Hoth, which informed the game's focus on overwhelming enemy pressure and positional defence.

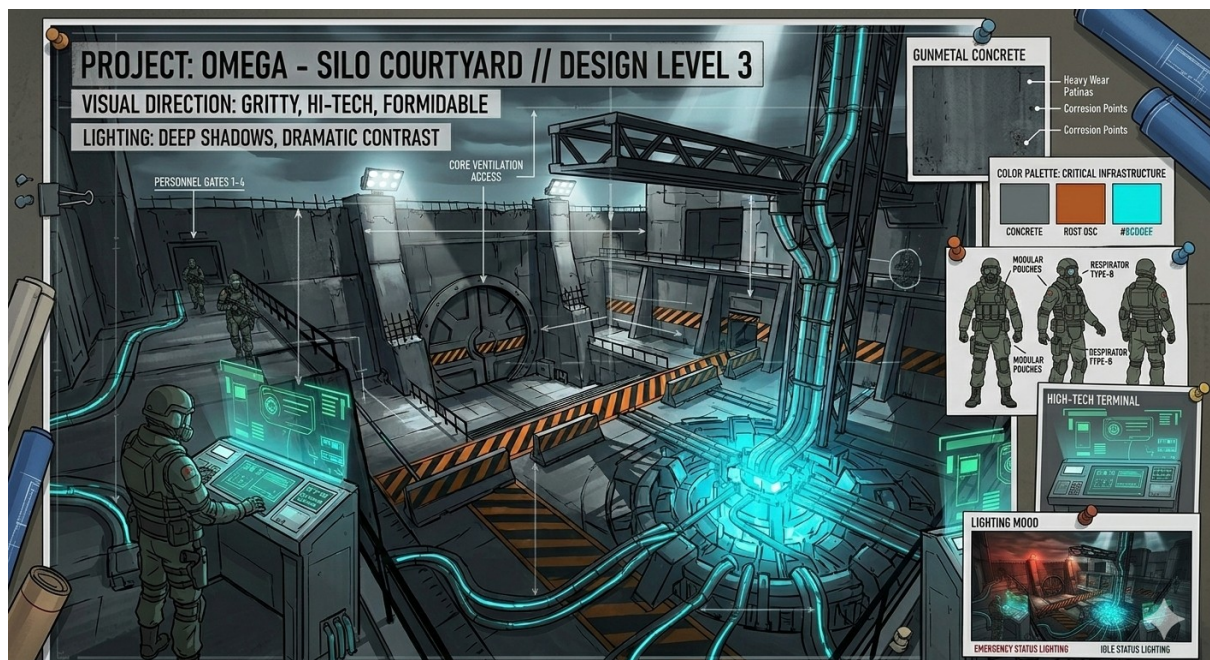
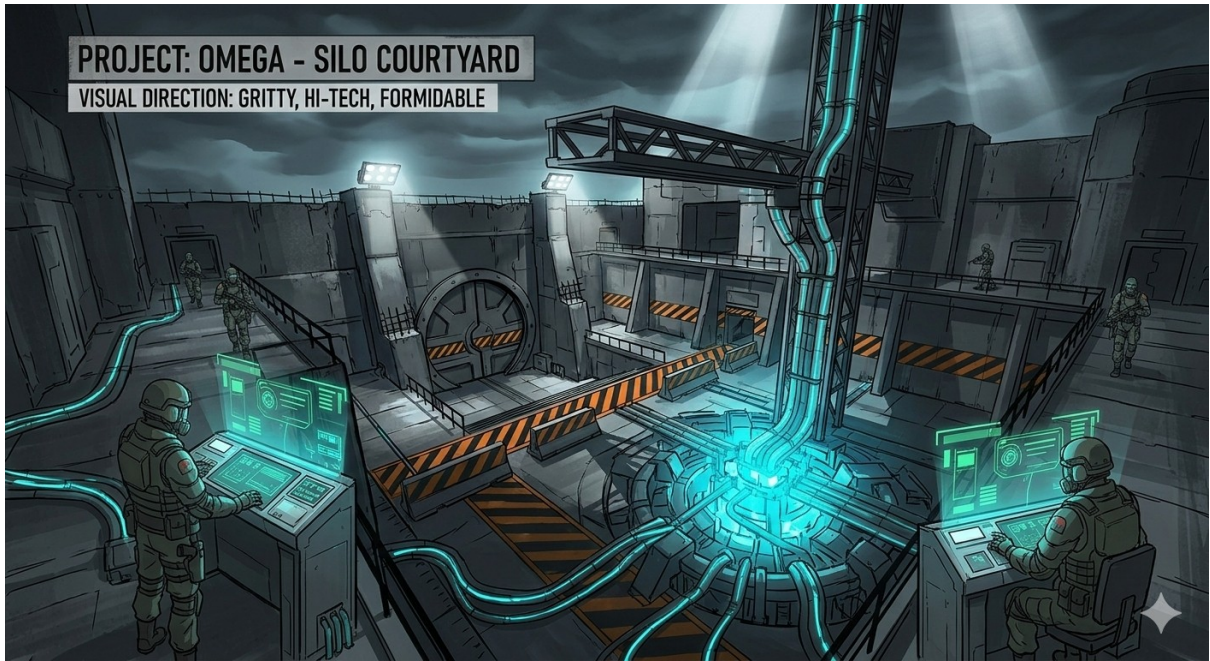


Figure 1: Map Design inspiration concept

PROJECT: OMEGA - SILO COURTYARD

VISUAL DIRECTION: GRITTY, HI-TECH, FORMIDABLE



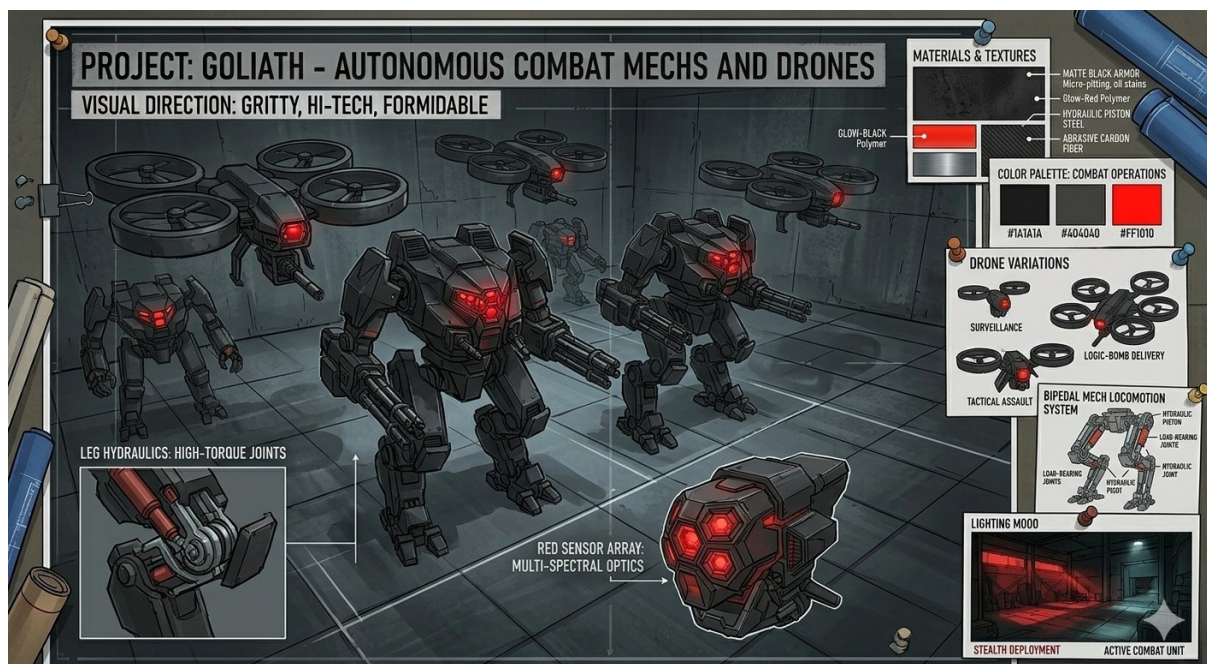
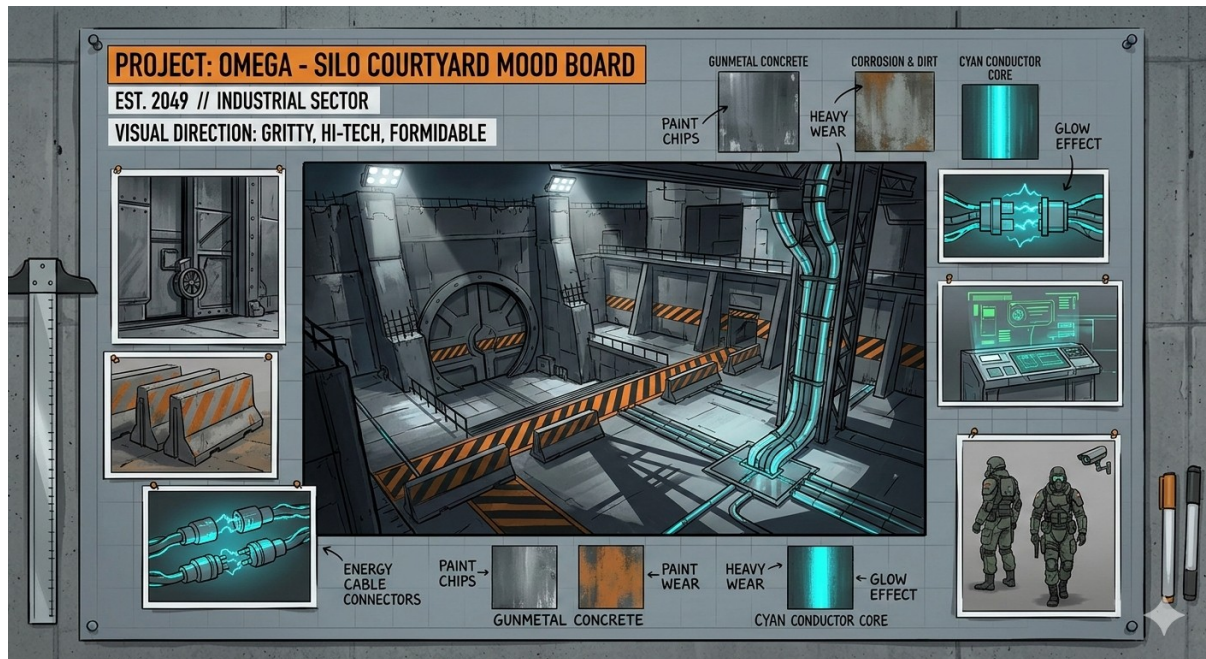


Figure 2: Mech and Drones concept



Figure 3: Hell Diver's defense continuous wave inspiration

Hell divers defense missions

Star wars Battle Of Hoth defending against the Empire's walkers etc



Figure 4: Battle Of Hoth Turret Inspiration

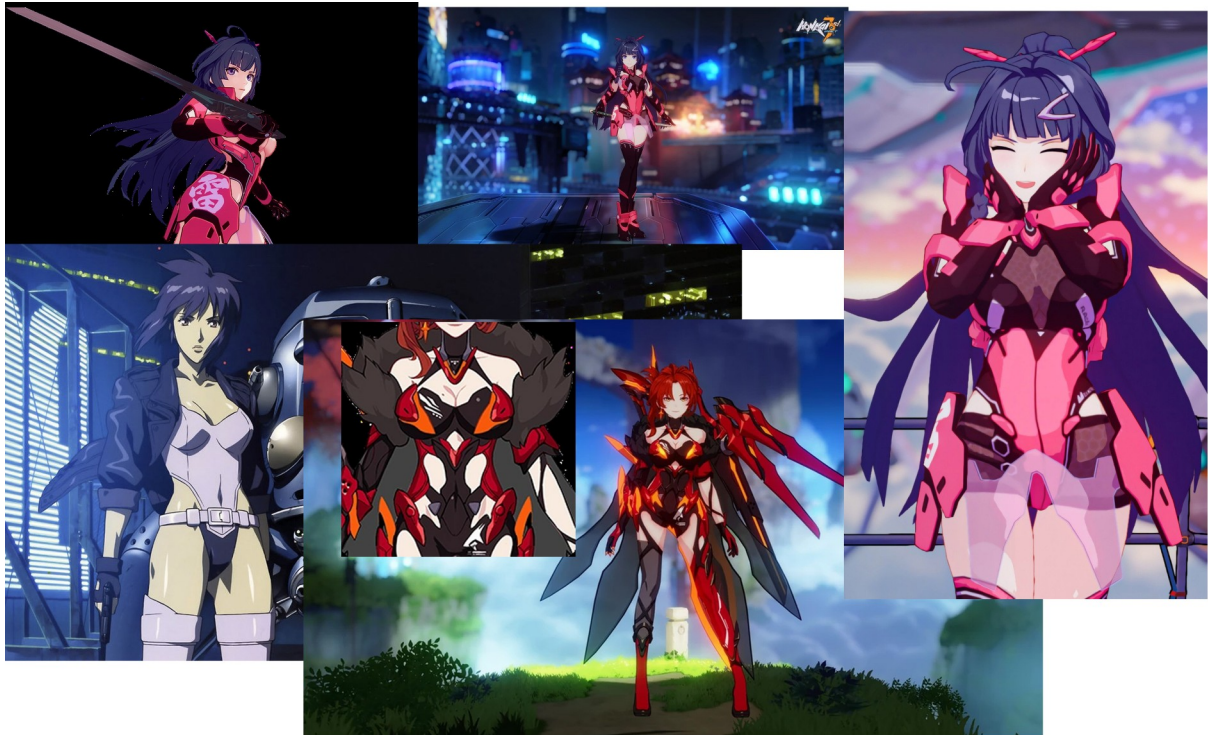


Figure 5: Honkai Impact + Ghost in the Shell Mood board



Figure 6: Nier Automata - Setting inspiration

Colour Scheme and Visual Identity

The game employs a clear friend–foe colour coding system, with blue used for player projectiles and red for enemy attacks. This ensures immediate visual readability during high-intensity gameplay, supporting rapid decision-making under pressure.

The overall aesthetic draws inspiration from NieR: Automata and Honkai Impact 3rd, blending futuristic sci-fi elements with a gritty, industrial war-zone atmosphere. This hybrid style reinforces the game’s tone of a desperate, high-stakes last stand.

Enemy mech design plays a key role in maintaining visual clarity. Mechs utilise a dark, desaturated metallic palette (charcoal and black tones), contrasted with bright red illuminated components, such as optics. These glowing elements act as immediate indicators of hostility, naturally drawing player attention during combat. Their sharp, angular silhouettes and heavy mechanical limbs further communicate aggression and reinforce their role as militarised threats.

In contrast, Annie’s design balances functional sci-fi armour with a stylised anime influence. Her primary palette consists of charcoal and silver tones accented with deep red highlights, creating a cohesive “elite operative” identity that is both grounded and visually distinctive.

Subtle Valkyrie-inspired motifs, including winged iconography, are integrated into the design to support world-building and suggest her role as a final defender.

An alternative blue and gold palette was explored, evoking a more heroic and noble tone. However, this approach suggested a “regal protector” archetype better suited to a pristine, high-tech setting. The final silver and red palette more effectively aligns with the industrial environment, reinforcing Annie’s identity as a resilient survivor within a hostile battlefield.



Figure 7: Character Design - Colours



Figure 8: Alternative Character Colour Palette



Figure 9: Mech Design

Gameplay and Core Mechanics

Objectives and Progression

The primary objective in *Annie's Last Stand* is to successfully initiate a nuclear launch by defending four active generators, which collectively contribute to the launch sequence progress. The game unfolds as a continuous combat experience, replacing traditional wave-based progression with sustained enemy pressure.

Each generator passively contributes to the launch meter over time. If all generators remain operational long enough, the launch sequence completes and the player achieves victory. However, if all generators are destroyed, the mission fails.

Gameplay is structured around a real-time combat–strategy loop, where the player must constantly balance positioning, threat prioritisation, and resource management under pressure.

Core Challenges and Player Interaction

The player controls Annie using 360-degree movement and aiming, engaging enemies in a high-intensity, bullet-heavy environment. Challenges arise from managing multiple simultaneous threats across the play space:

- **Swarm Mechs:** Fast, low-durability units that overwhelm defensive positions
- **Siege Mechs:** High-durability enemies that deal significant damage to generators
- **Aerial Drones:** Bypass ground defences, forcing repositioning or anti-air investment

Because enemies approach from multiple angles and target different generators, the player must constantly reposition, creating sustained tension and spatial decision-making.

Rewards and Penalties

- **Reward:** Destroyed enemies (“mech eliminations”) drop Scrap, used for upgrades, repairs, and defensive deployment
- **Penalty:** Generators take damage when enemies are not intercepted; loss of all generators results in failure

This creates a direct relationship between player performance and long-term survival, reinforcing urgency.

Objects and Systems

- **Annie (Player):** Defined by movement speed, health, and fire rate; primary agent of defence
 - **Generators (Objectives):** Four stationary structures that generate launch progress; must be protected
 - **Turrets:** Deployable automated defences that assist with area control
 - **Walls:** Defensive barriers that delay or redirect ground enemies
 - **Scrap:** Resource currency obtained through mech eliminations
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Core Rules and Spatial Dynamics

Enemies spawn from the lower regions and edges of the map, advancing toward generator positions. The player must intercept and manage threats before they reach critical structures.

A key challenge is multi-point defence: with four generators active simultaneously, players cannot be everywhere at once. This forces prioritisation, strategic movement, and effective use of deployable defences.

Gameplay Refinement Notes

- Feedback indicated that the original **wave-based system reduced tension**, as players experienced downtime between rounds
- Suggested shift toward a **continuous enemy spawning system** to maintain pressure and better align with the “Desperate Last Stand” PX goal
- Feedback noted that defending a single silo lacked spatial complexity
- Proposed adding **multiple objectives (generators)** to encourage player movement and prioritisation.

Design Changes Made

- Replaced wave system with **continuous enemy pressure**
- Introduced **four generator objectives** instead of a single silo health pool
- Adjusted enemy spawn positions to create **multi-directional threat vectors**
- Increased player decision-making by forcing **multi-point defence**

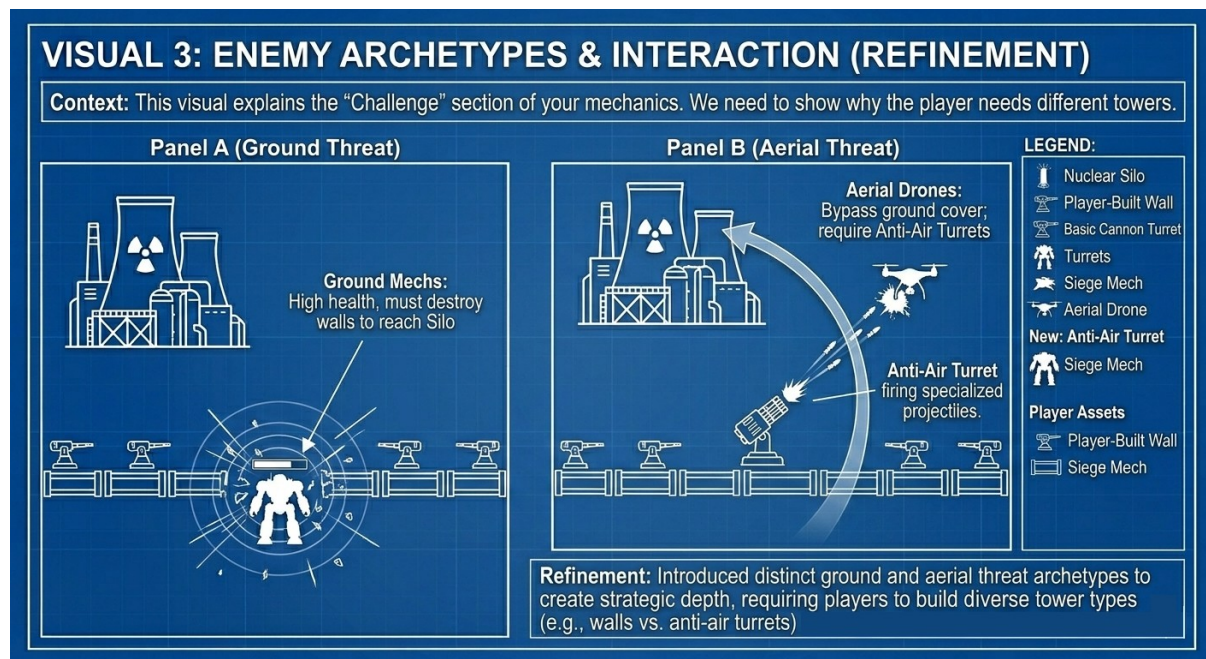


Figure 10: Enemy interaction concept

Development Roadmap

The following roadmap outlines the staged implementation of core gameplay systems, progressing from foundational mechanics to complete gameplay functionality.

Phase 1: Core Environment and Player Setup

The initial phase establishes the core play space and player functionality. A top-down arena is created, including the placement of the four generators as primary objectives. The player character (Annie) is implemented with 8-directional movement, alongside a static orthographic camera that follows the player.

Phase 2: Core Combat Mechanics

This phase focuses on implementing the core SHMUP systems. Mouse tracking is used to control aiming direction, allowing the player to rotate and face the cursor. A projectile system is introduced through a bullet prefab and pooling manager. Various weapons are introduced to fit different gameplay styles and gradual weapon upgrades.

Phase 3: Enemy Design and Behaviour

Enemy prefabs are developed with distinct roles:

- **Swarm Mechs (Basic bullet hell mechs and drones):** Low durability units
- **Siege Mechs (Tanks and Gunships):** High durability units that prioritise generator damage
- **Aerial Drones:** Flying enemies that can fly over the wall

Reinforcing enemy drop ships fly in at the bottom of the map, supporting a continuous spawning system that maintains constant gameplay pressure rather than discrete waves.

Phase 4: Core Systems and Economy

A resource system is introduced through Scrap, which is dropped upon mech elimination. Players collect Scrap via proximity-based interaction. Core damage systems are implemented, ensuring that enemies damage generators and the player, while player attacks reduce enemy health.

Phase 5: Defensive Systems and Shop Integration

This phase introduces strategic depth through deployable defences and upgrades. A shop system is implemented as an in-world interaction near the generators, allowing players to spend Scrap on turrets, walls, and upgrades.

Placement systems are refined to ensure valid positioning, and turret behaviour is implemented using detection ranges and automated targeting logic.

Phase 6: User Interface and Game States

The final phase focuses on player feedback and game flow. The HUD is implemented to display key information, including player health, generator status, Scrap count, and launch progress.

Interface / Visual System Details

The game features a simple screen flow: **Main Menu → Gameplay → Victory/Failure Screen.**

The in-game HUD provides real-time feedback, including player health/utility, Generator integrity, Scrap count, game progression, score. A shop interface is accessible near the silo, allowing players to purchase upgrades without leaving the play space, reinforcing risk–reward decision-making.

Visually, the game adopts a **2D top-down Isometric perspective**, combining flat environmental layouts with billboarded character sprites. This ensures clarity while maintaining stylistic depth.

The control system is standard for SHMUP design:

- Movement: WASD
- Aiming: Mouse
- Shooting: Left click
- Interaction: E key
- Inventory & UI toggles: Keyboard shortcuts

This control scheme prioritises responsiveness and precision, supporting high-intensity gameplay.

Control System

The control scheme is designed for responsiveness and accessibility, supporting precise movement and fast-paced combat.

Core Controls

- Movement: WASD (8-directional movement)
- Aim: Mouse cursor (determines firing direction)
- Shoot: Left Mouse Button

Interaction & Gameplay

- Interact (Shop / Repair / Objects / Pick up): E
- Place Turrets / Traps: Number keys (1–9 hotbar) or right click > place to activate the contextual placement system
- Remove Selected Item and place back in inventory: 0

Interface Controls

- Inventory Toggle: I
- Toggle Hotbar Visibility: H
- Minimap: M

System Controls

- Pause / Options Menu: ESC

Interface Design Notes

- Initial HUD concept included too many elements, creating visual clutter during combat
- Feedback recommended prioritising:
 - Player health
 - Generator status
 - Scrap count to remain in the inventory
 - Launch progress
- Decision made to place shop physically near objectives to introduce risk–reward tension

UI Refinements

- Simplified HUD layout to focus on critical gameplay information
- Added clear generator indicators (Alive/Destroyed status)
- Maintained consistent colour coding for projectiles (blue = player, red = enemies)

Game Help System

The game features a paginated Help System accessible from the Main Menu or optionally triggered when initiating a new game. This system utilises a sequence of text and image panels to clearly explain core controls, threat prioritisation, and the scrap economy. Additionally, it integrates brief narrative dialogue to establish the high-stakes setting of defending the final nuclear silo, ensuring players understand both their mechanical objectives and the context of their mission before combat begins.

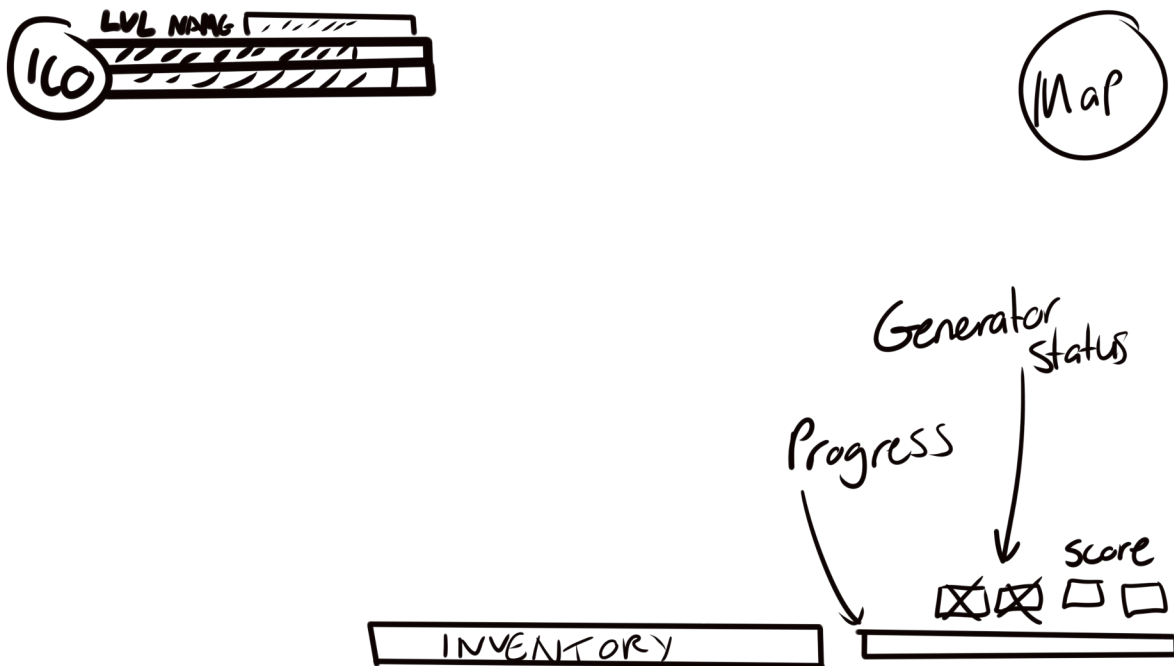


Figure 11: UI Design

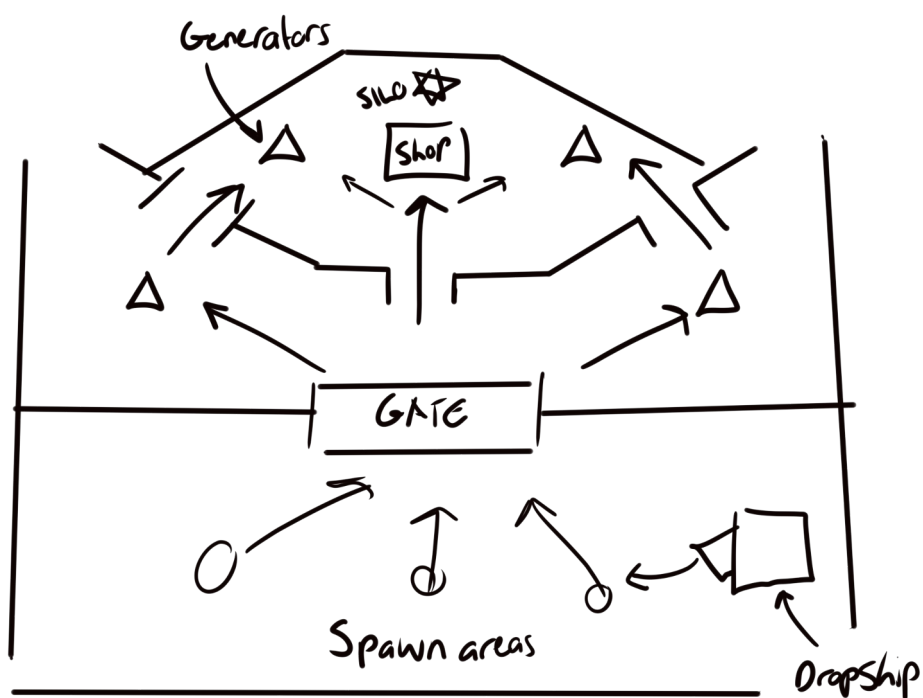


Figure 12: Gameplay diagram

Logo:



Figure 13: Logo

The logo reflects the game's gritty, industrial aesthetic through the use of bold shapes and a restrained colour palette. It features a stylised red eagle, representing Annie's organisation and reinforcing its presence within the game world.

The eagle motif connects directly to the subtle Valkyrie-inspired iconography present in Annie's armour, creating visual consistency across character and branding elements. The use of red emphasises themes of conflict, urgency, and militarisation, while also aligning with the established enemy colour language, reinforcing the hostile environment in which the game takes place.

Overall, the logo functions as both a narrative symbol and a visual anchor, supporting the game's tone of a desperate last stand while maintaining cohesion with the broader visual identity.

Game Design Reflection

What Worked Well

The defensive choke point design successfully reinforced the intended player experience of tension. By funnelling enemies through key areas, players were forced to make high-stakes prioritisation decisions under pressure.

Additionally, the integration of the shop within the play space created meaningful risk–reward dynamics. Players had to disengage from combat to upgrade, reinforcing urgency and preventing downtime (Similar to MOBA like LOL and DOTA).

Areas for Improvement

The initial implementation of specialised turret targeting (air vs ground) created unnecessary frustration, limiting player agency. Revising this system to allow more flexible targeting improved player satisfaction and strategic consistency.

Similarly, the original wave-based structure disrupted gameplay flow. Transitioning toward a more continuous enemy pressure model better supported the intended “bullet hell” experience and sustained tension.

Late-Game Difficulty and Player Mastery

A key aspect of the design is the intentional escalation of difficulty in the later stages of progression. At approximately 75% completion, tank-type enemies are introduced, subverting player expectations by targeting generators from previously safe positions behind the defensive wall. This moment is designed as a deliberate disruption, challenging assumptions established earlier in the game and reinforcing the need for adaptive strategy, such as defending them with higher HP defensive turrets.

The experience is not intended to be mastered on a first playthrough. Instead, the design encourages iterative learning, where players gradually develop an understanding of enemy behaviours, optimal weapon choices, and effective defensive configurations. This aligns with a mastery-based progression model, where failure becomes a critical component of learning rather than a purely negative outcome.

Difficulty scaling intensifies further beyond 90% progression, where the final stretch is intentionally disproportionate in challenge. The last 10% of gameplay is designed to feel significantly longer and more demanding, with tank enemies capable of rapidly dismantling player infrastructure, including multiple generators to slow down their launch progression. This creates a high-pressure endgame scenario that tests the player’s accumulated knowledge, adaptability, and prioritisation under extreme conditions.

Core Motivational Drivers in Player Engagement

The game targets four primary motivational drivers: character appeal, combat intensity, mastery, and progression.

Character appeal is intentionally incorporated through the design of the player character, presented as an attractive, anime-styled figure with a balance of visual appeal and practical armour. This serves as an immediate engagement hook, making the game more eye-catching while also fostering player attachment and identity. The character’s design encourages continued play by creating a stronger personal connection between the player and their in-game presence.

Combat intensity is expressed through the core gameplay loop, where players are required to continuously engage and destroy large numbers of enemies. The sustained pressure

reinforces a fast-paced, high-adrenaline experience aligned with the intended “bullet hell” design.

Mastery is developed through repeated play, requiring players to learn enemy behaviours, optimise defensive placement, and adapt to escalating threats—particularly with the introduction of late-game enemy types such as tanks.

Progression is delivered through the gradual acquisition of stronger weapons and defensive systems. This provides a clear sense of growth, enabling players to handle increasing difficulty while reinforcing long-term engagement.

Effective Design Processes

Player Experience mapping was critical in identifying mismatches between intended and actual gameplay feel, particularly regarding pacing.

Iterative prototyping allowed for rapid refinement of core mechanics such as choke points and enemy flow, demonstrating how small structural changes significantly impact player experience.

Feedback Discussion Notes

- Responses were positive to the continuous combat system, noting that it maintained constant engagement
 - Multi-generator system was identified as a strong mechanic for creating player-driven prioritisation
 - Some feedback highlighted that enemy roles needed clearer distinction during gameplay
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Identified Strengths

- The shift to continuous spawning successfully enhanced tension and urgency, aligning with intended PX goals
 - The introduction of multiple generators created meaningful spatial decision-making, preventing static gameplay
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Identified Issues

- Early turret designs felt overly restrictive due to limited targeting types, reducing player agency
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Iterative Changes Based on Feedback

- Adjusted turret systems to allow more flexible targeting behaviour
 - Reinforced visual clarity through stronger colour contrast and simplified UI elements
 - Refined enemy behaviour to better communicate their roles (e.g., clearer movement patterns)
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Reflection on Design Process

- Iterative prototyping was essential in identifying issues with pacing and player flow
- Player Experience (PX) mapping helped highlight mismatches between intended and actual gameplay feel
- Learned that removing systems (waves) can sometimes improve the experience more than adding complexity